FinTechathon 2026

Research and Project Roadmap

International Track · Preparation for a Bangladeshi student team

Historical research from 12 October 2026; contest sources and project evidence rechecked on 15 October 2026 for **Tideway**. This report covers the competition rules, previous winners, project choice, technical design, testing, submission and final-round preparation in one place.

> **Recommendation.** Choose **Topic E: SME Finance Copilot** and build **Tideway**, an approval-gated cash-flow rescue agent for small businesses. (Earlier working title: Probaho.) It finds a cash gap, plans a recovery with a risk-aware optimiser, negotiates with simulated counterparties and, after human approval and a signed mandate, completes the action inside a sandbox. There is no guarantee of winning; the aim is a tested project that matches the published judging criteria closely.

Current scope. The team supplied five public member names, all at BUET, in tracked `submission-team.public.json`. Leader, roles, nationality, eligibility, registration and submission route remain unconfirmed. The contest upgrades are local working changes; the current verification status belongs in `docs/READINESS_CURRENT.md`. Recording is explicitly deferred.

> **Most urgent now.** The deadline is **20 October 2026, 21:59 Bangladesh time**. Our own target is to submit by 20:00 on 19 October. Confirm registration first; registration may close early if seats fill. [S1–S3]

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1. Competition and eligibility

Name: 2026 Shenzhen International FinTech Competition, known as FinTechathon. 2026 is the eighth edition and the first with a separate International Track. Theme: "AI as a Financial Participant", meaning you must build an AI agent that can complete tasks in a defined financial situation. [S1–S3]

Who can take part

Condition	What we must do
Full-time students	Bachelor's, master's or PhD; current enrolment at a recognised eligible institution must be confirmed.
Institution location	Outside Mainland China, Hong Kong SAR, Macao SAR and Chinese Taipei. Students in Bangladesh are eligible if they meet the other conditions.
Team size	2 to 5 members; members may come from different departments, years, universities and countries. Solo submissions are not allowed.
At least 60% non-Chinese nationals	Teams of 2 need 2; of 3 need 2; of 4 need 3; of 5 need 3 (rounded up to a whole number).
One team only	A person cannot be in more than one team across the Domestic and International tracks.
Language	Submission, presentation and review are in English. A Bangla UI could be a bonus, but a judgeable English flow is essential.

Organiser naming mismatch

The CMU announcement names Tsinghua PBC School of Finance and WeBank as hosts. The IEEE CIS Singapore page lists Shenzhen University WeBank Institute of Financial Technology as host and a Tsinghua research institute as technical guidance institution. Do not paraphrase organiser names in the application; use the prescribed form. [S1, S3]

Four team decisions before taking part

1. Verify every member's enrolment and nationality.
2. Does at least one member have the means and a passport to travel to Shenzhen for the final?
3. Does everyone have the right to use the code and data?
4. One team lead controls the application, revisions and the deadline.

> The short rules do not give detailed permission for AI coding tools. Do not assume that "independently completed" either bans or allows all AI use. Ask the organiser for a written answer, and keep a record of the tools used and the human contribution. [S2]

2. Timeline, prizes and how to apply

Stage	Published date	Our decision
Registration opens	31 August 2026	Finalise the roster now.
Registration and submission close	20 October, 23:59 Beijing (UTC+8)	Bangladesh: 20 October, 21:59.
Compliance screening	21 to 22 October	Detailed schedule from IEEE.
Online review	23 to 28 October	CMU gives a wider window, 20 to 31 October.
Finalists announced	31 October	IEEE: invitation letters the same day.
Check-in and WeBank visit	20 November	Learn the travel approval and booking rules first.
Grand Final and awards	21 to 23 November	At least one member must attend in person.

Dates are based on [S1–S3]. In the original ITS web announcement the intermediate dates are shifted by one step; the attached rules and the IEEE schedule are more consistent. Treat any later written update from the organiser as the latest instruction. **Dhaka is two hours behind Beijing** (UTC+6 against UTC+8). Do not use the London, Tokyo or Sydney conversion examples in the PDF; they contain conversion errors.

International Track prizes

Prize	Amount
First (1)	RMB 100,000
Second (1)	RMB 80,000
Third (1)	RMB 50,000
Best Innovation (1)	RMB 20,000
Excellence (several)	Certificates

Total RMB 250,000, all before tax. Prizes go to the team, not to each member. There is no registration fee. The organiser pays for or arranges round-trip international travel and accommodation for invited finalist members; the invitation states how many members. Do not assume that visa fees, meals, local transport or the costs of all members are covered. [S2]

Where to apply

CMU and the attached PBCSF rules: wangyr@pbcsf.tsinghua.edu.cn; CMU asks for a completed form and each member's passport bio page. IEEE Singapore: fintechathon@ican-x.com, with a confirmation reply promised within 48 hours. Confirm in writing now which route applies to your team. Do not send passports to several addresses unnecessarily. [S1–S3]

IEEE registration subject: `International Track Registration – Bangladesh – [Team Name] – [Team Leader's Full Name]`. Save the confirmation or team ID; do not hold back registration until the project is finished.

3. What to submit and how it is judged

The list below comes from pages 2 to 4 of PBCSF Appendix 2 as published on ITS. It is a competition-branded rules document; the full terms on the official website and later notices also apply. [S2]

Material	What we will prepare
Technical documentation	Architecture, core algorithm and security design, with data flow and limitations.
Presentation deck	A deck suitable for a presentation of at most 10 minutes. No fixed slide count was found.
Demo video	Optional on a separate checklist, at most 5 minutes, but requested under execution evidence, recording remains deferred until the team instructs it.
Source code	GitHub repository link, README, deployment instructions and test cases. Confirm repository visibility.
Security self-assessment	How permission tiers are enforced, and a known-risk list.
Execution evidence	A to E: operation logs from a simulated account or sandbox, and the demo video. F: real on-chain transaction hashes.

Published scoring

Category	Weight	Our evidence
Task completion	40%	A workflow that runs to the end; tool logs; sandbox state before and after.
Security and compliance	30%	Permission enforcement, tenant isolation, attack tests, self-assessment.
Innovation and interaction	30%	Accuracy of the problem, comparable alternatives, easy explanation, real user feedback.

Topic F has a 0 to 5 on-chain exploration challenge bonus; the core rubric is the same. The exact way the bonus is added and its cap are unclear. Choosing a wallet only for the bonus could damage the core 100 points. Whether testnet transactions are acceptable must be confirmed in writing. [S2]

Our submission package

`README.md; docs/architecture.pdf; docs/security-assessment.pdf; slides/pitch.pdf; demo video; evidence/run-log.jsonl; evidence/results.csv; tests/; LICENSE; THIRD_PARTY_NOTICES; .env.example`. These are our proposed names, not mandatory organiser names. No secrets, passports or private customer data in the repository.

The latest submission before the deadline prevails, and changes after it are said not to be accepted. Save a snapshot and the commit SHA at the deadline. Whether bug fixes or enhancements are allowed before the final must be asked separately. [S2]

4. Previous winners and lessons

The 2026 International Track is new. The projects below won in other tracks of earlier FinTechathons; they show the kind of preparation involved but are not a confirmed 2026 winning formula.

Year and prize	Team or institution	Project
2025 AI, 1st	HKUST Guangzhou DSA team	Ling Cao Pan: natural language to backtested, executable quantitative strategies. [S5]
2025 AI, 2nd	AI4Fin; Capital University of Economics and Business	Financial service combining an LLM, a data foundation and intelligent investment. [S6]
2025 AI, 3rd	one piece; HKUST Guangzhou	FinAgent: finance and investment assistant based on bill data. [S5, S6]
2025 AI, creativity	Sio team; Nanjing University	EchoPolis: Gen Z financial-behaviour insight with AI digital twins. [S6]
2025 Data, 1st	Anti-fraud team; Peking University	Scalable mixture-of-experts ensemble for anti-money-laundering and fraud. [S6]
2025 Data, 2nd	Guard Nine Five; BNU Zhuhai	Microenterprise credit scoring from multidimensional data. [S6]
2025 Data, 3rd	Retention; Shenzhen π and Lanzhou University	Digital-bank customer churn modelling with TabNet and LightGBM. [S6]
2025 Data, creativity	Finfluence; Shenzhen University	Social influence and the banking-product ecosystem. [S6]
2024 Blockchain, 1st	Beijing Institute of Technology	Trusted supervision and transaction traceability for AIGC models. [S7]
2024 Blockchain, 2nd and 3rd	HKUST Guangzhou; Tianjin University	CipherBridge encrypted computation; AIGC lifecycle copyright management. [S7]
2023 AI, 1st	Aa100K; five-university team	OncoFed: tumour diagnosis based on FATE. [S8]
2023 Blockchain and Product, 1st	BUPT; Beihang–CUC–ESCP team	A14C cultural digital products; conversational WeBank digital steward. [S8]
2022 AI and Blockchain, 1st	Fed-healthcare; TIDAL	Federated body-sound perception; fine-grained blockchain data permissions. [S9]

The 2025 AI champion team: Wang Hao, Wang Liping, Yu Haolin, Zhu Xinyi; supervisor Chen Lei. FinAgent members: Peng Guanrun and Zhang Chengtan. Latin spellings are transliterated from the published Chinese names; follow the original institutional record in any application or contact. [S5]

> **Our lesson.** Show a complete workflow, verifiable execution, privacy and security, and a specific user problem together. The champions had research and pilots before the competition; rather than imitate that, we build a small but robust workflow. Full name lists for all tracks in 2019 to 2021 and 2024 were not confirmed here.

5. Comparing six ideas

These scores are our own decision aids, not the official rubric and not a probability of winning. Each is rated 1 to 5. Weights: build in the time available 30%, task evidence 25%, security demonstrability 20%, differentiation 15%, user and data access 10%.

Topic and idea	Time	Task	Security	Differentiation	Access	Total /5
A Expense Recovery Assistant	5	4	4	2	5	4.10
B Explainable Wealth Agent	2	3	2	2	2	2.25
C Student Tuition Planner	4	4	4	4	4	4.00
D Claim Evidence Assistant	4	4	4	3	3	3.75
E Tideway SME Copilot	4	5	5	4	4	**4.45**
F Policy-Bound Wallet	2	5	2	4	2	3.05

First choice: Topic E

A computable cash gap, several alternative actions and permissioned sandbox execution can show evidence in all three parts of the 40/30/30 rubric. If we can speak directly with local shops, agencies or wholesalers, user validation also becomes easier. This is a possibility, not confirmed data access.

Second choice: Topic C

If we cannot find SME users within 24 hours but can find 3 to 5 international students, "TuitionBridge" is the fallback: a tuition payment plan that reacts to fee quotations, deadlines and FX changes, with a sandbox payment instruction after approval. Real bank or remittance integrations take time; rates and fees need timestamps.

Status of the other ideas

Topic A is quick but risks becoming a generic expense chatbot. D traces policy clauses to a claims checklist well in a demo, but needs insurance documents and domain review. For B, suitability and investment correctness are hard to prove. F involves much work on wallet security, signing and transaction evidence; reconsider it only if the team already has blockchain skills.

Honesty about novelty

QuickBooks already offers AI cash-flow guidance and invoice reminders; Float offers scenario planning. So we will not claim to be the "first AI cash-flow assistant". Our proposed difference, in one narrow SME workflow: evidence-linked cash rescue even with little data, explicit payroll and reserve constraints, a risk-aware optimiser, negotiation, enforcement of execution permission outside the model, and replayable safety tests. This is our product hypothesis, not a claim to be the only one in the market. [S10, S11]

The name "Tideway" is a working title; trademark and domain availability have not been checked. (The earlier Bangla working title, Probaho, meant "flow".)

6. Chosen project and the user's problem

Tideway: an approval-gated cash-flow rescue agent for small businesses. First user: a small service business or wholesaler with regular invoices, supplier payments and payroll. The first version has one currency per business and one main workflow, and ships with three synthetic businesses in three countries (Bangladesh in BDT, China in CNY, Singapore in SGD). New businesses sign up as their own workspace, in any of 29 currencies.

> **Core question.** "Can I pay salaries on Friday? If not, which bill should be pushed back, which customer should be asked to pay early, and what will happen if customers pay late?" The agent calculates the gap, compares feasible plans, negotiates with simulated counterparties and, after human approval, runs the approved action in the sandbox.

Step	What the user sees	What the system does
1 Import data	Overview and records; CSV import	Validates transactions, invoices and bills completely before writing; quarantines suspicious text.
2 Understand risk	Cash weather, three scenarios, "N of 20 futures fall short"	Exact daily cash projection with a 14-day look-ahead; customer payment profiles; seeded 1,000-run Monte Carlo.
3 Plan	Recommended bundle and alternatives with cost, risk before and after	Exact optimiser over deferrals, early-payment offers and reminders; mock financing compared, never executed.
4 Approve	Review and approve, PIN, typed amount	Maker-checker; two people above the threshold; HMAC-signed, expiring mandate.
5 Negotiate	Accept, counter-offer or reject per counterparty	Simulated counterparties; a counter becomes a new proposal; a refusal triggers a re-plan.
6 Execute	Receipt with before/after state	Idempotent, atomic execution; state hash and read-back verification.
7 Prove	Audit chain, exports, tamper drill	Hash-chained audit with witness checkpoint; evidence JSON, CSV and draft ISO 20022 pain.001 (never sent).

Demo numbers (Shapla Services, synthetic BDT)

Opening cash 120,000; payroll 80,000; supplier bill 70,000; confirmed inflow 20,000. Scheduled minimum cash on 31 October = 120,000 + 20,000 − 80,000 − 70,000 = **−10,000**. With a minimum reserve of 20,000 the reserve shortfall is **30,000**. The optimiser then finds the cheapest combination of deferrals and early-payment offers that restores the reserve on every day of the horizon and checks it against simulated customer lateness. Every plan shows its assumptions; a reminder is never counted as cash collected; a deferral only takes effect after the (simulated) supplier accepts.

Skills we will show separately

- A verifiable action plan from an ambiguous financial request; plans that break a constraint are rejected.
- Execution with permission; recovery after a failed call (same idempotency key returns the same receipt).
- An honest "no safe plan" outcome with escalation.
- Every number from the agent traced to a deterministic calculation and a source record.

Out of scope

Real-money transfers, production bank integration, automatic loan approval, stock trading, a full ERP, mixed currencies inside one business. The financing module is a clearly labelled mock catalogue comparison, not a promise of credit.

7. Technical design and data

Design: English web UI → HTTP server (role, CSRF, rate limits) → agent and service layer → deterministic projection, analytics and planner → approval and mandate store → sandbox execution → hash-chained audit. The model, if used, plans and explains; the backend is the only authority for money, balances and permission.

Part	Implementation	Acceptance condition
UI	Vanilla JavaScript single-page app: landing page, sign-up and join-by-code, then Overview, Rescue, Assistant, Inbox, What-if, Records, Team, Security and Evidence; no build step	Full flow works on fixtures; screenshots captured by script.
Backend	Python 3.11 standard library HTTP server; loopback only	Strict field validation; per-business isolation.
Calculation	Integer minor units; exact decimal parsing; daily ledger	Exact arithmetic tests; mixed currencies rejected.
Analytics	TidePay lateness model (simulated FedAvg, illustrative masks) with credibility-weighted profiles as fallback; seeded Monte Carlo (1,000 runs); counterparty risk map	Same records and seed give identical results.
Planner	Exact search over at most one option per record; Monte Carlo re-check of the best 40	Reserve held every day; payroll funded; reasons for every rejected record.
Counterparties	Deterministic simulated suppliers and customers	Accept, counter or reject; labelled simulated.
Storage	SQLite: plans, approvals, mandates, executions, audit, sessions	Atomic transactions; execution history survives reset.
Security core	PBKDF2 PINs, HMAC-SHA256 mandates, token-bucket rate limits	Self-test cases pass.
Evidence	results.csv, run-log.jsonl, verification-summary.json, screenshots	Generated by `scripts/evaluate.py`.

Data contract

Transactions: id, date, amount_minor, currency, type, category. Invoices: id, customer, amount_minor, due_date, expected_collection_date, status, paid_date. Bills: id, payee, payee_account, amount_minor, due_date, priority, reschedule_allowed. Policy: reserve_floor, action_limit, daily_cap, dual_approval_threshold, look-ahead days, autonomy level. Amounts are positive integers in minor units; the transaction type gives the direction.

Cash calculation and risk

Closing cash each day = previous day's cash + that day's inflow − that day's outflow, in three scenarios: scheduled, likely (each customer's expected delay) and stress (a high-percentile delay). Each open invoice's delay comes from TidePay's predicted distribution (likely = expected delay, stress = 80th percentile); the credibility blend of own history and the business-wide average (weight n/(n+5)) remains the explainable fallback. The Monte Carlo draws one delay per customer per simulated future and reports the probability of breaching the reserve, negative cash and unfunded payroll. We do not claim statistical forecast accuracy; the back-test is on synthetic history. Early-payment discounts are priced as an annual rate d/(1−d) × 365/days (2% for 20 days ≈ 37%). Financing comparisons disclose fee, term, repayment and eligibility assumptions.

Agent and tools

Tideway's own AI. *TidePay* predicts each open invoice's lateness band and is trained by simulated federated averaging across 40 synthetic SMEs in one process with illustrative, reproducible masks. The process holds the data and pools training features; no deployed privacy guarantee is claimed (log loss 1.036 against 1.55 for the old rule). *TideLM* is Qwen3-4B-Instruct with a LoRA adapter we trained with MLX on a laptop: it turns an owner's

message in eight languages into a typed tool plan and turns pasted bank, card and mobile-money messages into draft records while flagging injection. On held-out examples with unseen names the deployed v2 adapter raised exact plans from 26.7% to 96.7% and extraction F1 from 0.834 to 0.969 (`evidence/ml/tidelm_report.md`). It runs on the owner's machine; plans pass a typed validator, replies are written by code from tool output, and the model has no money or write tool. Answers come in the language of the question (English, Chinese, Spanish, French, Portuguese, Indonesian), with the same numbers.

Agent tools: `read_ledger`, `profile_customers`, `project_cash`, `simulate_risk`, `check_security`, `compare_plans`. There is no approve or execute tool, no raw SQL, shell or arbitrary HTTP. Each request has at most eight tool calls. The default local agent needs no network. The optional LLM mode (own API key, per-message consent) sends only pseudonymised names and tool output, and flags any number in the answer that no tool produced.

Demo dataset: three synthetic businesses (Shapla Services in BDT, Lotus Print Studio in CNY, Harbourline Traders in SGD), each with settled history and open invoices and bills, designed so each tells a different story: a payroll gap, hidden risk from customer habits, and a case with no safe plan plus planted attacks. Synthetic and real results are never mixed.

8. Security and permission model

Security and compliance are 30% of the official rubric. The tiers and controls below are our implementation, not an organiser-published tier numbering, and not a production compliance certification.

Tier	Who	Can
Read	Agent	Read records, project cash, simulate, analyse.
Propose	Agent	Create plan proposals and message drafts; no external mutation.
Approve	Owner or accountant	Approve a bundle with PIN step-up; two people above the dual-approval threshold.
Execute	Owner or accountant	Execute a signed, accepted mandate once.
Delegated execute	Agent under a standing mandate	Zero-cash reminders only, five a day.
Deny	Everyone	Real money movement, credit decisions, payroll deferral, acting on unverified bank details. The model cannot override these.

Approval is more than a button

Approval requires a fresh PIN. The signed mandate binds the actions, amounts, counterparties, plan and snapshot hashes, policy version and a 15-minute expiry. If any field changes, the old approval is void. Before execution the system re-verifies the signature, expiry, the human or standing-mandate principal, that every counterparty accepted, and that the underlying records have not changed. The same idempotency key returns the same receipt; a second key on a used mandate is refused.

Controls

- **Kill switch and circuit breaker:** pause stops planning and execution; only the owner resumes, with a reason. Five security denials in ten minutes pause the agent automatically.
- **Payee fraud:** changed bank details and look-alike payee names block all actions on the bill. Only flagged payees can be verified: owner-only, by callback to a number on file. Whoever verified cannot be the only approver for that payee (separation of duties).
- **Injection:** hidden Unicode is stripped and flagged; instruction-like text already in ledger records is quarantined as data and excluded from planning/model context. Inbox text is read locally only to draft records, then deterministically screened and held for human confirmation.
- **Autonomy dial:** Advise, Propose or Delegate; moving money is never delegated.
- **Web layer:** loopback only, Host and Origin checks, CSRF token, HttpOnly SameSite=Strict cookie, strict CSP, rate limits, spreadsheet-safe CSV export.

Privacy and credentials

No real passports or merchant identities in the repository. Minimum fields only; names are pseudonymised before any optional external model call and record free text is never sent. Credentials live in a local `.env` that is never committed.

Failure recovery and audit

Do not treat a timeout as failure and retry blindly: query by idempotency key first. Execution is atomic. A pending mandate can be cancelled; an executed action is not shown as "undone" (a reversal would be a separate transaction). The audit log is a hash chain with an independent witness co-signature, so edits and truncation are detected; a one-click drill demonstrates this on a safe copy. A hash chain alone does not protect against a fully privileged attacker.

Security report and test map

The security assessment contains assets and trust boundaries, the permission matrix, a threat table mapping each of the 30 self-test case ids to OWASP LLM and Agentic identifiers, known risks and residual limitations. It does not claim certification against any bank or jurisdiction.

9. Testing and evidence

Targets below are separated from achieved results. Development data and the final holdout must be kept separate; raw evidence must exist for every number we report.

Test	Set and target	Status
Task completion	End-to-end scenarios; target at least 90%	**Done**: 5/5 task cases. Versioned broader set: 24 distinct conditions, 12 development and 12 held out; outcome classes reported separately.
Money arithmetic	Exact minor-unit results	**Done**: unit tests, including the Shapla −1,000,000 minor on 31 Oct and 3,000,000 shortfall.
Permission safety	Forbidden-action attempts; zero unauthorised mutation	**Done**: 13 authorisation, mandate, kill-switch and delegation attacks blocked.
Retry and replay	Duplicate and restart cases; at most one mutation per action	**Done**: replay and double-spend cases.
Injection and fraud	Malicious input; zero attack success	**Done**: 3 injection and 4 payee-fraud cases.
Usability	3 to 5 SME owners or domain reviewers, with consent	Planned; not yet collected.
On-device model quality	TideLM vs base model on 500 held-out examples with unseen names	**Done**: historical deployed v2 exact plans 26.7% to 96.7%, extraction F1 0.834 to 0.969, injection-flag accuracy 91.5% to 98.5%. Synthetic templates with unseen names, not real-world accuracy. Changed-runtime device inference remains unverified.
Latency and cost	Median per answer on a 16 GB laptop; no per-call cost (on-device)	Historical device measurement: about 5 s per TideLM answer before prompt caching; changed-runtime latency is unverified; small talk answers instantly. Cloud-LLM runs: not measured.

Total today: **30/30 self-test cases pass** (5 task, 25 attack), plus the automated unit, HTTP, conversation, language, SaaS and regression suite (latest count in `evidence/test-summary.json`). Reproduce with `python3 -m unittest discover -s tests -v` and `python3 -m tideway.selftest`.

Historical comparison plan and new measured evidence

The preserved narrow experiment compares one fixed deferral rule with the optimiser on 20 opening-cash perturbations (10/20 versus 12/20 feasible). The new suite has 24 independently authored business conditions; 24/24 outcome handling, including 9 executed rescues, not 24 rescues. A generic-LLM quality comparison was not run. See docs/SCENARIO_EVALUATION_V2.md.

Failure cases to cover

1. Duplicate invoice
2. Missing currency
3. Negative opening balance
4. Supplier refusal or no consent
5. Late receivable
6. Payroll and reserve conflict
7. Invalid model output
8. Timeout after execution
9. Expired mandate
10. Another business's object id
11. Changed payee
12. Prompt injection in an uploaded CSV

Evidence record and claims

Per run: case id, category, OWASP mapping, pass or fail, elapsed time and observed outcome (`evidence/run-log.jsonl`, `results.csv`); a summary in `verification-summary.json`. We do not store hidden chain-of-thought; a short rationale and source evidence suffice.

> We may say "30 of our 30 defined sandbox cases pass", if true. We may not make broad claims from a small synthetic benchmark such as "cash flow improves 40%" or "fraud is fully stopped". If time saved is measured on a small sample, state the sample size, task and method next to the number.

> **Release gate:** if any unauthorised mutation or money-calculation error exists, disable action execution in the submission demo and fix it first. We will not hide broken security for points.

10. Plan for 15 to 20 October 2026

Times are Bangladesh time. The core and local upgrades are implemented; source changes remain local until the team reviews and commits them. This plan covers verification, feedback and submission, with video recording deferred. Work can be shared, but keep one integration owner. AI-generated code must not be merged without human review and tests.

Date	Main work	Verifiable result at day end
Thu 15 Oct	Registration check; roster and eligibility; send the organiser question list; re-run all tests on a second machine; re-generate evidence.	Registration confirmation or written route; 30/30 self-test and the automated test suite (count in evidence/test-summary.json) reproduced.
Fri 16 Oct	Baseline comparison (rule, generic LLM, Tideway) on the three fixtures plus new scenarios written before tuning; arrange 3 to 5 SME or finance-reviewer sessions with consent.	Baseline CSV; interview schedule.
Sat 17 Oct	Run interviews and usability sessions; independent attempt to break the sandbox by a teammate; fix only real bugs; **scope freeze**.	Findings list, fixes, frozen commit.
Sun 18 Oct	Rehearse the live linked-delay demo; finish the deck; regenerate PDFs; clean-machine run from the README. Recording remains deferred.	Live-demo rehearsal, `slides/pitch.pdf`, architecture and security PDFs; video remains outstanding.
Mon 19 Oct	Final package check; confirm links from a reviewer account; **submit by 20:00**; save receipt, SHA and archive hash.	Submission and receipt saved.
Tue 20 Oct	Access check; fix only if the submission is broken.	Final update target 18:00; hard cutoff 21:59.

Team roles

The five confirmed public names and BUET institution are in submission-team.public.json. Actual contribution roles and leader must be supplied by the team; no role assignments are inferred here.

Smaller scope if time is short

Import, deterministic cash timeline, one recovery plan, approval, sandbox schedule change and audit log. Freeze optional feature expansion and use the labelled deterministic fallback when needed; retain tested multilingual support. Show one clean happy path and one blocked malicious request correctly.

Go or no-go checkpoints

- End of 15 October: if no eligible team or registration route is confirmed, resolve that before anything else.
- End of 17 October: if baseline or security findings are still open, stop adding features.
- End of 18 October: if documents or critical verification are unfinished, freeze features; resolve deferred video requirements separately.

Whether this is possible depends on the team's skills; do not hide lack of time behind a long feature list.

11. Pitch, demo and judges' questions

Plan for a video of at most five minutes

Time	On screen
0:00–0:25	Real problem: payroll approaching, cash short; storm cash weather.
0:25–1:15	Projection, customer profiles, Monte Carlo risk, source rows.
1:15–2:20	Rescue agent run; recommended bundle with cost and risk before and after; mock financing.
2:20–3:50	Dual approval with PIN and typed amount; signed mandate; negotiation (accept, counter, reject); execution receipt and read-back.
3:50–4:20	Blocked changed-payee, look-alike and injected records; honest "no safe plan".
4:20–4:40	Self-test 30/30, tamper drill, kill switch; limits and next steps.

Ten-minute pitch

Slides: problem; user problem and target; workflow; live demo; architecture; security; evaluation; existing alternatives; business model; limitations and roadmap. Leave enough time for the financial demo. Ten minutes is the maximum in the rules; confirm whether Q&A is separate.

Likely judge questions

- **"Why is this different from a general chatbot?"** An exact cash engine and optimiser, backend policy enforcement, stateful execution; baseline on the same test set once it is run.
- **"What if the AI is wrong?"** It cannot act. Figures come from tools; answers are grounding-checked; execution needs a signed mandate; refusal and clarification paths exist.
- **"Why is all the data synthetic?"** Reproducibility and attack coverage; state it plainly; give any small consented pilot separately.
- **"Where is the novelty?"** Not a generic forecast; the contribution is a constraint-aware plan through to safe action, with enforcement outside the model. We do not claim a novel research algorithm.
- **"No real bank integration?"** The rules accept sandbox evidence for A to E; we show the adapter boundary (a draft ISO 20022 file that is never sent). No production-readiness claim. [S2]
- **"Who would pay?"** Small business owners or accountants; a subscription hypothesis; say that pricing is not validated.
- **"Is the mandate the AP2 standard?"** It is modelled on the AP2 idea (verifiable, scoped, expiring authorisation); it is not an implementation of the specification.

Demo backup: fixed seed fixtures, a reset path, PDF screenshots and an automatic fresh-database rehearsal, and fallback if a model times out. Never present a recording as live execution.

12. Final preparation and cost

The first milestone is a valid submission; then finalist selection. Do not plan on updating the project for November without written confirmation of the post-deadline change policy. [S2]

Phase	Work
21 to 30 October	Keep the submitted commit unchanged; Q&A rehearsal; gather provenance and licence evidence.
31 October	Finalist notice, invitation letter, and how many members are funded.
1 to 7 November	Visa process, university approval, attendance and flights per organiser instructions.
8 to 14 November	Bug fixes or pilot if allowed; otherwise rehearse the submitted version.
15 to 19 November	English rehearsal, deployment access, offline backup, travel documents.
20 November	Check-in and WeBank visit.
21 to 23 November	Final presentation; have code and test evidence ready.

Prototype budget range

These are not vendor quotes; they are our internal spending caps. Model API BDT 3,000 to 7,000 (only if the optional LLM mode is evaluated); hosting and storage 0 to 2,000; interview, mobile data and incidentals 1,000 to 2,000; contingency 1,000 to 2,000. Total roughly BDT 5,000 to 13,000, excluding existing laptops and team time. Do not assume free credits; set a daily usage cap. Run 10 workflows first and measure cost per run; budget = expected runs × measured cost per run + hosting + buffer. Paid tools are not required for video editing or slides. The exchange rate and net amount of any prize are not estimated here.

Clarify before travel

How many tickets and accommodations; direct booking or reimbursement; airfare cap; airport transfer; visa fee; meals; insurance; arrival and departure dates. No non-refundable booking at our own cost without written confirmation. An invitation letter does not guarantee a visa.

Value after the competition

Win or not, the reusable outcomes are working code, a security test suite, audit design, user interviews and a short product demo. Before any real deployment, a real bank contract, jurisdiction-specific legal review, stronger security review and data-processing agreements would be needed; these are not deliverables of this demo.

Business-model test: show the workflow to 5 SME owners and 2 accountants; ask what they would pay for, how they do it today, and what stops them uploading data. Do not manufacture market size or revenue from interviews.

13. Submission checklist and open questions

Tick every box

☐ 2 to 5 eligible members; enrolment and nationality correct.
☐ Registration confirmation; no member is in another team.
☐ Topic E and the project title (Tideway) are the same everywhere.
☐ English documents, deck and narration; video ≤ 5 minutes.
☐ The README runs in a clean environment.
☐ Judges can open the repository, demo and video links.
☐ Architecture, algorithm, permission design and known risks are included.
☐ Test cases, fixture licence, execution log and receipts are included.
☐ Every reported number matches raw evidence.
☐ Secrets and personal data are removed.
☐ Code dependencies and third-party attributions are included.
☐ Final commit SHA, zip hash, submission timestamp and email receipt are saved.

Questions to send the organiser today

1. Which is the authoritative registration and submission email for Bangladesh-based teams?
2. Is registration still open, and how do we get a confirmed slot?
3. What is the correct secure way to submit the form with passport and enrolment proof?
4. May we use AI coding assistants and our own previously written code? What disclosure is needed?
5. Is a self-built sandbox sufficient for topics A to E, and is there any mandatory dataset or API?
6. The video is optional on the checklist but required in execution evidence: will submitting a video remove the ambiguity?
7. What are the rules for a private GitHub repository, file size, document template and link access?
8. After 20 October, are bug fixes or enhancements allowed for finalist demos?
9. For F, are testnets acceptable and how is the bonus counted?
10. How many finalists, how many minutes of Q&A, and how many travellers are covered?

When our decision would change

Keep E if at least 3 SME stakeholders are available and the core coding skills exist. If SME access is missing but student-finance users are available, an early pivot to C is possible. Changing topic after 15 October costs a lot of time; reducing scope is usually better. Reconsider F only if the team already built secure wallet infrastructure.

Personal information

Five team names and BUET were supplied publicly and tracked; years, passport status, roles, leader, coding experience and budget have not been invented. Once verified, roles and scope can be made more specific. No registration or email was sent as part of this research.

14. Sources and limits of verification

Original research date: 12 October 2026; the four contest sources were re-opened on 15 October 2026 (docs/CONTEST_SOURCES_CHECKED.md). For rules, the competition-branded Appendix, the university announcement and the technical-guidance institution's page were cross-checked; where they disagree, the relevant section says so.

- S1 CMU announcement, 7 October 2026: ird.oou.cmu.ac.th/2026-shenzhen-international-fintech-competition
- S2 PBCSF Appendix 2 (p. 2 rubric, p. 3 submission, p. 4 final and IP): its.ac.id PBCSF Appendix 2
- S3 IEEE CIS Singapore, 2026 International Track: ieee-cis-singapore.org/2026-fintechathon
- S4 Official competition website: fintechathon.g-ican.com
- S5 HKUST Guangzhou, 2025 AI winners: hkust-gz.edu.cn
- S6 InfoQ, 2025 results and image listing the eight awards: infoq.cn/article/cKMpwF1wq9Q9MtQSGR5d
- S7 FISCO, 2024 blockchain winners: fisco.com.cn/en/news_7/862.html
- S8 Shenzhen University, 2023 winners: szu.edu.cn/info/1161/18428.htm
- S9 FinTechathon 2022 results: infoq.cn/zones/fintechathon/campus2022/result
- S10 Intuit, cash-flow agent capabilities: intuit.com blog
- S11 Float, scenario planning: floatapp.com/blog/scenario-planning-cash-management
- S12 LangGraph workflow design documentation: docs.langchain.com
- S13 CMU registration form: drive.scmc.cmu.ac.th

Confirmed facts versus recommendations

The deadline, rubric and submission list are facts from the published rules. The budget, future acceptance targets, schedule and idea scores are our proposals, not measured outcomes or organiser approval. Current implemented features and finite measured results are separated in Sections 15 and 16 and the versioned evidence reports.

The dynamic content of the main website and the full Privacy Policy could not be verified. The fields in the CMU form download were not fully verified. An older English asset contained terms from a different competition; these were not used as 2026 International Track rules. No assumption fills those gaps. A separate enterprise competition with the same name exists; its winners are not included as FinTechathon student-track winners. Selected, verifiable results for 2022 to 2025 were taken; a complete archive of every prize in every year is not claimed.

15. Contest upgrade: evidence and live state change

Reproduce with `python3 scripts/evaluate_tidepay_v2.py`, `python3 scripts/evaluate_upgrade_scenarios.py` and `python3 scripts/demo_payment_delay.py`. Reports and raw outputs are in `evidence/upgrade-v1/`; historical observations remain preserved.

Prediction evidence (audit benchmark, not shipped weights)

Train/validation/test: 7,745 / 1,433 / 2,376 invoices, 40 synthetic businesses. Global chronological cutoffs and payment-label maturity prevent unavailable labels crossing training boundaries. All fitted statistics use training data; observable rolling history is rebuilt as of invoice issue.

Same 2,376 test invoices	MAE days	Log loss	Sum Brier
Prior	11.9856	1.5882	0.7936
Heuristic	7.4107	1.5688	0.7905
Local-only	7.2152	1.2329	0.6407
Simulated federated	6.9568	1.0727	0.5777
Pooled	6.9448	1.0927	0.5855

Federated MAE is worse for 7/40 businesses; the pooled MAE is slightly better overall. The historical prior used test-label mean statistics; its raw report is retained separately. Reliability bins, counts, business-cluster intervals and split hashes accompany the new report. This is one synthetic cohort, not production accuracy or a cryptographic federation guarantee.

Workflow and screening

24/24 defined outcomes handled: 9 executed rescues, 9 honest infeasible, 2 no-action and 4 validation/payee blocked. Only 7/9 rescues meet the 10% simulated-risk target. Injection screening detects 12/14 labelled attacks and flags 7/14 benign messages. Payee screening detects 6/8 attacks and flags 3/8 benign cases; a case-only account-history bypass was fixed with the initial result preserved. Finite tests are not proof of security. Performance details and residual failures are in docs/SCENARIO_EVALUATION_V2.md.

Live financial decision

Approve the original Shapla proposal. As Nadia, explicitly apply Bay Logistics' collection date of 15 November through the separate owner/PIN demo action; What-if stays read-only. The new snapshot invalidates old authority and a stale execution attempt is refused. The fresh proposal replaces supplier deferral with early-payment offers; Bay's 1.5% counter requires fresh approval before negotiated execution and verified read-back. Use a fresh unused database filename for each full rehearsal; never erase earlier evidence.

Remaining verification and administration

The changed model-tool runtime is bounded, traced and tested with injected outputs; native MLX inference remains outstanding. Section 16 records the later actual compatible Qwen3 Mac challenge, a separate model path. Python/macOS/Chrome checks do not establish Windows, Docker or another device success. Recording, eligibility, registration, route confirmation, leader/roles and reviewer access remain team-controlled. No push, publication, submission, email or video recording is included in this work.

16. Historical review: exact evidence and approval gaps

Current index: `docs/READINESS_CURRENT.md`. Earlier sections describe their dated experiments. The prototype remains within synthetic SME cash-flow planning, human approval and sandbox execution. No bank, lender or real-money rail is connected.

Exact shipped TidePay artifact

Frozen weights were evaluated without fitting on 7,427 new synthetic invoices from 80 new SMEs and four seeds. Shipped-artifact MAE is 8.5585 days, versus 10.2296 for the customer-history heuristic and 11.7453 for a historical-training prior. Log loss is 1.2850 and sum Brier 0.6712. The original 2,376-record chronological audit above evaluates different weights. No-history log loss is worse than the training prior; runtime/training history-order differences are disclosed. This is synthetic evidence, not real-world accuracy.

First unseen screening and actual native Mac inference

The independent sealed screening set detected 2/24 attacks and flagged 2/24 benign messages. Payee intent detected 10/12 attacks with 4/12 benign flags; the observable callback policy handled 24/24 cases. Unrelated impostors cannot reliably be distinguished from genuine new vendors using these fields alone. A missing warning is not a trust decision.

Pinned Qwen3-4B Q4_K_M with llama.cpp b11678 was provisioned and verified on macOS arm64. The first 32-request challenge handled 28 correctly, with 23/24 relevant tool/argument cases and 5/8 safe stops; all 32 produced no money execution, mandate or financial-ledger mutation. Median request latency was 1.225 seconds. Amount/term confusion, unknown-name substitution, short-name expansion and incomplete conditional recovery were retained, then addressed with separate deterministic guard regressions. Repeating known examples is not a fresh blind result. Native MLX adapter and receipt-model evaluation remain outstanding.

Approval, repayment and release

Every above-target approver must explicitly accept stored residual risk. The server binds acknowledgement to plan ID/hash, bundle, probability and target, and retains it in audit, signed mandate and receipt. Stale or altered consent is rejected. Mock financing compares arrival, fee, total repayment, APR convention and unverified eligibility, then projects through repayment so borrowing cannot hide a later shortfall.

`python3 scripts/prepare_release.py` verifies one frozen LF source tree, tests desktop/mobile workflows, rebuilds five PDFs, creates the local ZIP and checks fresh extraction without Git or site packages. It neither downloads models nor submits anything. An anonymous consent-aware SME study kit is ready; actual participation is zero. GitHub-hosted CI, Docker, external security review and consented user feedback remain unverified. Team administration and video remain deferred.

17. Historical judge and auditor follow-up

On 17 October 2026, the Appendix and IEEE guidance were reopened and the latest checkout was reviewed afresh. The untouched Windows baseline had one failure and one error in 252 tests. Repairs address signing-key replacement and concurrent creation, malformed signature exceptions, the ignored structured-plan setting, future payment history in snapshot analytics, reused mock collateral, LF artifact bytes and a SQLite test handle leak.

Nine added behavioural regressions bring the suite to 261 cases. Current counts, skips and exact source hashes belong in `evidence/test-summary.json`. A separate actual Windows compatible Qwen3 run handled 32/32 previously disclosed requests with 32/32 containment. All 34 generations and historical first observations are retained; this is a known-case regression, not blind accuracy. The current browser evidence also distinguishes actual inference from injected outputs.

`docs/INDEPENDENT_AUDIT.md` maps requirements to evidence and lists fixed and unresolved findings. The coding assistant's review is not an external penetration test or organiser approval. The unfamiliar-message detector still caught only 2/24 attacks and flagged 2/24 benign messages in its first unseen challenge. There are zero real SME study participants; native MLX/receipt quality, hosted CI and Docker remain unverified. Team details, registration, route, reviewer access, timed rehearsal and the deferred video require the team's actual actions.

18. Owner constraints and current local release

Current implementation and results: `docs/REMEDIATION_V7.md` and `docs/READINESS_CURRENT.md`. Earlier test counts and verification gaps above belong to their original observations; stored hosted-CI and Docker receipts do not attest to this local remediation.

A 1% owner discount ceiling rejects a 1.5% counter-offer. Successor proposals preserve the ceiling, preference, exclusions and target; they need fresh approval and exact counterparty consent. Store validation, signed mandates, receipts and audit retain the constraints. Legacy proposals without recorded instructions need a fresh owner-reviewed proposal.

H11 reproduces approximately 11.78% breach risk over 30,000 documented draws against the 10% target. Likely-path feasibility is separate from risk-target attainment. Every required approver must acknowledge exact above-target risk; this does not guarantee a rescue.

Native screening uses the already cached Qwen3 base without the planning LoRA. Disclosed development, failed prompt runs and the independently authored sealed fresh evaluation stay separate under `evidence/screening-v7/`. Raw outputs, model hashes, confusion matrices, uncertainty and latency are retained. Classification never authorizes payments or removes independent verification. No model weights were downloaded.

The final local package binds current tests, financial scenarios, browser checks and extracted startup to one frozen source digest. Exact counts and hashes are in the generated evidence and `dist/` receipts. Independent remediation review agents were unavailable because of an account usage limit. No independent review pass, publication or submission is claimed.